

Assignment 2 (30 Marks)

Scientific Machine Learning: Blending Science with Data

Question 1 (8 marks)

You are given that a physical model for a particular engineering system is given by

$$\frac{d^2y}{dx^2} + py(x) = \cos(x) \quad (1)$$

The domain length is $x \in [0, 2]$ and the boundary conditions are $dy/dx(x = 0) = 0$ and $y(x = 2) = 5$. p is constant.

- a) Using finite difference schemes, write down the discretized equations of the equations above.
- b) Write a Julia program that solves the resulting discretized equations and plot the solution for $p = 4$. Note the value of $y(x = 0)$ for this value of p .
- c) In this problem, you are allowed to change the value of p . Find the value of p that makes $y(x = 0) = 1$. For this part of the question you will need to write your own gradient based optimisation technique taught in lectures (i.e. not allowed to use the Julia packages for optimisation()). You are allowed to use any of the automatic differentiation packages in Julia to find gradients. Document all your steps and explain clearly how you arrive at your solution.
- d) Check your answer in c) with any of the optimisation packages in Julia

Question 2 (10 marks)

You are given temperature measurements on a 1m plate with thermal conductivity, k . This plate is being heated with an unknown heat flux. You are provided with 4 data files with temperature values at location x

- $k = 0.1$ ('Data01.csv')
- $k = 0.5$ ('Data02.csv')
- $k = 0.3$ ('Data03.csv')

- $k = 1.5$ ('Data04.csv')

The first two are your training data set and the last two would be your testing dataset.

- Plot the data for $k = 0.1$ to see how they look like.
- Train a Neural Network that models all the data from only the first ($k=0.1$) dataset. See if you can use this model to predict all the datasets (i.e. for all k). Your Neural Network must have 2 inputs (x and k) and 1 output ($T(x, k)$). It can have as many hidden layers as you like and you can use any activation function.
- Train a Neural Network with just the ($k = 0.1$ and $k = 0.5$ (training) dataset. See how accurately this model predict all the datasets.

Question 3 (12 marks)

You know that the governing equation of temperature on a plate is governed by the following governing equation

$$k \frac{d^2 T}{dx^2} = -q(x) \quad (2)$$

You are given temperature data (taken via thermocouples) from a 1 m rod. The rod has temperature 5°C at $x = 0$ and the rod is insulated at $x = 1\text{m}$, i.e $dT/dx = 0$ at $x = 1$.

- Use Neural Network to model $q(x)$, solve Eq. (2) and optimise the coefficients of the Neural Network such that the predicted T is as close as possible for the data given to you for $k = 0.1$. Your Neural Network must have 1 input, x , and 1 output ($q(x)$). It can have as many hidden layers as you like and you can use any activation function.
- Plot the approximated $q(x)$ predicted by the Neural Network.
- Use this model for $q(x)$ to predict the temperature for $k = 0.5, 0.3$ and 1.5 . Is this a good model?